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Corporate green innovation under environmental regulation: The role of ESG ratings and greenwashing

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ABSTRACT

China's economic trajectory has transitioned from prioritizing rapid growth to emphasizing quality, efficiency, and sustainability. In this context, balancing environmental protection with the promotion of a green economy has become increasingly significant. This study investigates the relationship between environmental regulation and green innovation in Chinese enterprises using data from listed companies spanning 2007 to 2019. The results indicate that environmental regulations positively influence green innovation among firms, a finding validated by robustness checks, including instrumental variable methods. Furthermore, the study identifies corporate greenwashing behaviors and environmental, social, and governance (ESG) performance as mediating factors. Firms resort to greenwashing strategies to mitigate regulatory pressures while simultaneously fulfilling their ESG responsibilities, thus contributing to green innovation. Additionally, the study reveals that the impact of environmental regulations on green innovation is heterogeneous, varying according to the external economic environment, firm ownership, and industry characteristics. The positive effect is more pronounced under lower levels of government intervention, in private enterprises, and among energy and new energy sectors. These findings underscore the nuanced dynamics between regulatory frameworks and corporate innovation in China.

1. Introduction

China's rapid industrialization and urbanization have yielded significant economic gains over recent decades, but these advancements have also led to severe environmental degradation. Between 2000 and 2019, China's carbon dioxide emissions surged nearly threefold, from 340 million tonnes to 1006.5 million tonnes, contributing nearly 30 % of global emissions. Additionally, the annual average PM2.5 concentration in China stood at 34 micrograms per cubic meter in 2019, far exceeding the World Health Organization's recommended levels. In response to these pressing environmental challenges, the Chinese government has escalated its regulatory efforts. The 13th Five-Year Plan for Ecological and Environmental Protection, released in 2017, targeted a reduction in carbon dioxide emissions per unit of GDP by over 18 % by 2020 (from 2015 levels), and aimed for more than 80 % of cities at the prefecture level and above to meet good air quality standards.

Corporate green innovation is seen as a key factor in advancing high-quality economic growth (Zhang et al., 2024; Zhang and Kong, 2022). However, there remains debate among scholars regarding the effects of environmental regulations on such innovation. Some argue, as Porter and Linde (1995) suggest, that stringent regulations can drive corporate innovation and enhance competitiveness. On the other hand, critics warn that regulations could lead to "greenwashing," where companies engage in superficial or misleading environmental practices to avoid compliance pressures (York and Rosa, 2003; (Kong et al., 2022)). Recent research highlights the nuanced impacts of environmental regulations. For example, Fang et al. (2021) found that the introduction of China's new Environmental Protection Law led to an increase in green patents by exerting pressure on highly polluting industries. Conversely, Wang et al. (2022a, b) identified an inverted U-shaped relationship, indicating that overly stringent regulations can stifle innovation. Additionally, Chen et al. (2022) argue that while market-based incentives and public engagement are effective in promoting green innovation, command-and-control approaches have limited impact. Given these mixed findings, it is crucial to further explore how environmental regulations can be optimally designed to encourage meaningful green innovation.

The relationship between environmental regulations and corporate innovation has been widely studied. Numerous scholars contend that such regulations can drive green technological advancements by

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enhancing corporate environmental responsibility and spurring increased R&D investment (Wang et al., 2021). Additionally, some researchers highlight the complex interplay between environmental regulations and economic policy uncertainty, which can sometimes hinder green innovation (Li et al., 2021). However, limited attention has been given to exploring the intricate dynamics between environmental regulations, corporate greenwashing practices, and green innovation, though this area is starting to gain traction. For instance, Kolcava (2022) found that while public accusations of greenwashing can erode trust in self-regulation, they have only a minimal impact on broader policy support. Similarly, Peng et al. (2021) revealed that stringent environmental regulations in nearby regions might accelerate the diffusion of green technologies and knowledge, thereby promoting green innovation, but the potential for greenwashing remains a concern.

Existing literature suggests that the interactions between environmental regulations, greenwashing, and green innovation are complex and warrant further investigation to uncover the specific mechanisms and conditions under which these relationships evolve. This study aims to fill this research gap by examining how environmental regulations, corporate greenwashing behaviors, and green innovation are interconnected in the Chinese context. Specifically, the study explores three main questions: First, how do environmental regulations influence corporate green innovation? Second, to what extent do companies resort to greenwashing as a strategy to avoid regulatory pressure? Third, how do environmental, social, and governance (ESG) ratings and greenwashing practices mediate the relationship between environmental regulations and green innovation?

To address these research questions, we analyze data from Chinese listed companies spanning 2007 to 2019 using a panel fixed-effects model to examine the relationship between environmental regulations and corporate green innovation. The results demonstrate that environmental regulations play a significant role in promoting green innovation among firms. However, the panel fixed-effects model is subject to potential endogeneity, policy influences, and random factors. For instance, regional economic development and the implementation of corporate environmental policies might correlate with green innovation outcomes. To mitigate these concerns, we first adopt the air circulation coefficient as an instrumental variable, following the approach of Hering and Poncet (2014), and employ a two-stage least squares regression analysis. Additionally, we conduct robustness checks using subsamples that exclude regions with stringent environmental regulations and those covering different time frames. Our findings remain consistent across these robustness checks, confirming that the results are resilient to issues of endogeneity, policy variations, and random factor interference. Moreover, we recognize the risk of selection bias due to the diverse industrial characteristics of firms, which could lead to biased estimates. Therefore, we focus on companies in highly polluting, energy-intensive, and overcapacity industries, and the robustness of our findings holds even after accounting for this sample adjustment.

This research contributes to the literature in two key ways. First, we provide evidence that environmental regulations positively influence corporate green innovation, challenging the perspective that such regulations might stifle innovation. Our findings emphasize the need for well-calibrated and context-sensitive environmental regulations to effectively drive green innovation. Second, we integrate environmental regulations, corporate greenwashing behaviors, and green innovation into a unified framework, using data from Chinese listed firms. Unlike previous studies that often focus on environmental policy, corporate social responsibility, or technological innovation in isolation, our research bridges these domains to address the dynamic interactions between environmental regulations and green innovation, thereby filling an important gap in the literature.

2. Research hypotheses development

2.1. Environmental regulations and corporate green innovation

In recent years, environmental regulation has emerged as a critical strategy for governments to steer businesses toward sustainable practices. Such regulations push companies to curtail pollutant emissions and enhance resource efficiency by setting rigorous environmental standards and policy guidelines. This external pressure compels firms to improve their environmental management systems while also motivating them to invest more in green innovation to achieve both environmental and economic objectives.

Environmental regulations can drive corporate green innovation through what is known as the “innovation compensation effect.” The Porter hypothesis suggests that stringent regulations can encourage firms to engage in innovative activities, resulting in processes and products that not only comply with environmental standards but also bolster competitiveness. For instance, companies subject to tighter emission controls may ramp up their investment in clean technology research and development (R&D) to reduce pollution control costs while boosting production efficiency. Additionally, such regulations heighten corporate awareness of environmental responsibility, prompting firms to innovate in order to manage compliance costs and mitigate environmental risks. Companies often take proactive steps, such as designing low-carbon products and refining production methods, to safeguard their market reputation and gain a competitive edge (Li et al., 2023). Moreover, environmental regulations can level the playing field by establishing uniform standards for all businesses, thereby promoting fair competition. This competitive environment encourages firms to pursue green innovation as a means to enhance their market position, creating a positive cycle of innovation and competition (Zhang and Wang, 2022). Based on these considerations, we propose the following hypothesis:

Hypothesis 1. Environmental regulations can promote corporate green innovation.

2.2. Environmental regulations, greenwashing, and corporate green innovation

Although environmental regulations can stimulate corporate green innovation, they may also lead to “greenwashing” behavior. Greenwashing refers to companies' deceptive or exaggerated promotion of their environmentally friendly measures to mislead consumers and investors into believing that the companies perform well in environmental protection. Although greenwashing is generally considered negative, its impact on green innovation is complex.

The stringent implementation of environmental regulations increases compliance costs and pressure on companies, forcing them to seek cost-reducing strategies. Greenwashing is a low-cost strategy that can temporarily alleviate economic pressure, allowing companies to allocate more resources to genuine green innovation. The market recognition and consumer trust that companies gain through greenwashing may secure additional capital and market opportunities, indirectly supporting green innovation (de Jong et al., 2017). Companies engaging in greenwashing may also face stricter regulatory and public scrutiny, which may drive them to increase their investment in green innovation in the long term to avoid reputational damage and legal risks from false claims.

Thus, while initially intended to evade stringent regulatory requirements, greenwashing can lead to external supervision and market pressure that ultimately incentivize genuine green innovation (Wang et al., 2020). Greenwashing can also serve as a transitional strategy for companies coping with regulatory uncertainty. As environmental regulation policies stabilize and mature, companies may shift toward genuine green innovation to maintain their competitive advantage. Therefore, greenwashing can be seen as a transitional strategy for green

innovation, providing a buffer period to adapt to new environmental regulations (Flowers et al., 2020). Based on the above analysis, we propose the following hypothesis:

Hypothesis 2. Environmental regulations may lead to corporate greenwashing behavior, but greenwashing can alleviate corporate burdens and promote green innovation.

2.3. Environmental regulations, ESG, and corporate green innovation

Driven by global sustainable development goals, environmental regulation has gained importance as a governmental tool that is crucial for controlling corporate environmental behavior. At the same time, corporate ESG ratings, as comprehensive indicators of corporate sustainable development performance, have become focal points for investors and stakeholders(Zhang et al., 2024). The combined pressure of environmental regulations and ESG ratings constitutes a significant force driving corporate green innovation.

ESG ratings offer a comprehensive system to evaluate companies, subjecting them to external supervision and assessment across each dimension. High ESG ratings both enhance corporate market image and competitiveness and attract further green investment. Companies must take practical actions in environmental protection, social responsibility, and corporate governance to achieve higher ESG ratings, further motivating them to engage in green innovation (Wang et al., 2022a, 2022b).

The interaction effect of environmental regulations and ESG ratings leads companies to adopt a more holistic strategy for sustainable development in response to environmental pressures. While fulfilling environmental responsibilities, companies must also consider social and governance factors, which encourages them to adopt comprehensive and systematic strategies for green innovation. For instance, when developing green products beyond environmental performance, companies must consider their social responsibilities and optimize their governance structure to improve their overall ESG rating. The combined pressure of environmental regulations and ESG ratings compels companies to actively pursue green innovation while fulfilling environmental, social, and governance responsibilities (Li et al., 2023). Based on the above analysis, we propose the following hypothesis:

Hypothesis 3. Environmental regulations exert pressure on companies to actively engage in fulfilling their ESG responsibilities, thereby driving green innovation.

3. Research design

3.1. Data source and processing

We analyze a sample of Chinese listed companies from 2009 to 2019 to examine the influence of environmental regulations on corporate green innovation. The data utilized in this study are primarily drawn from the CSMAR (China Stock Market & Accounting Research) database. Company patent data are sourced from the China National Intellectual Property Administration and the WIPO Green Patent List. Regional and city-level data are retrieved from the China City Statistical Yearbook, China Environmental Statistical Yearbook, and other relevant local yearbooks. The data processing steps are as follows: (i) Companies with incomplete or missing key information are excluded. (ii) Firms categorized as ST, ST*, or PT are removed from the sample. (iii) Financial industry firms, including those in securities, insurance, funds, and banking, are excluded. (iv) Companies that were not listed or became newly listed during the sample period are excluded based on their listing age. After these adjustments, our final sample comprises 3174 listed firms, resulting in 20,489 firm-year observations. To address the potential impact of outliers, we apply winsorization to continuous variables at the 1 % and 99 % levels.

3.2. Variable descriptions

(1) Dependent Variable

Corporate Green Innovation (*GreenPatent*): Consistent with the approaches used by Wang and Wang (2021) and Wang et al. (2023), corporate green innovation is quantified based on the number of green patent applications, which encompass both green invention patents and green utility model patents. Since some firms may not have green patent applications and the patent data are discrete, we apply the methodology of Fang and Hu (2023) by taking the natural logarithm of one plus the number of green patent applications to assess corporate green innovation.

(2) Core Independent Variable

Environmental Regulation (*EEFI*): Building on the approaches of Li and Tao (2012) and Ye et al. (2018), we measure environmental regulation by selecting pollutant emission indicators at the prefecture-level, including industrial wastewater discharge, SO_2 emissions, and smoke and dust emissions per unit of output value. These indicators are combined to create a comprehensive environmental regulation index using the entropy method. In this index, higher values reflect greater pollution and weaker regulatory enforcement, whereas lower values indicate stricter regulations. The detailed calculation procedure for the index is provided in Appendix A. To further validate our findings, we also incorporate the Baidu Index of Environmental Pollution (Ser) as an alternative indicator for environmental regulation robustness checks.

(3) Mechanism Variables

Corporate ESG Performance (*ESG*) and Corporate Greenwashing Score (*GreenWash*): First, the variable capturing ESG performance (*ESG*) measures the sustainability and social-value impact of corporate operations in terms of each of the three ESG dimensions. Following Xie and Lyu (2022), we use the quarterly Huazheng ESG rating (score from 1 to 9) averaged annually to measure corporate ESG performance. Second, following Walker and Fang (2012) and Huang et al. (2019, 2020), we construct a symbolic and substantive behavior indicator system for corporate environmental governance across four dimensions: governance and structure, process and control, input and output, and compliance (see Appendix B). The ratio of symbolic to substantive environmental governance behaviors is used to measure the corporate greenwashing score. The specific calculation steps are detailed in Appendix B.

(4) Control Variables

Referring to (Peng et al., 2023; Wang et al., 2024), we choose several control variables, such as firm size (*Size*), ownership structure (*Soe*), cash flow (*Cflow*), equity concentration (*Herf*), asset structure (*Stru*), return on assets (*Roa*), growth potential (*Growth*), and Tobin's Q (*TQ*). The variable information is shown in Table 1.

3.3. Model specification

To examine the impact of environmental regulation on corporate green innovation, we construct the panel fixed-effects model:

$$GreenPatent_{ijt} = \alpha_0 + \alpha_1 EEFI_{ijt} + \alpha_2 Control_{ijt} + \gamma_j + \lambda_t + \varepsilon_{ijt}, \quad (1)$$

where $GreenPatent_{ijt}$ is the level of corporate green innovation of firm i in industry j in year t; $EEFI_{ijt}$ represents the intensity of environmental regulation in the city in which the firm is located; *Control* represents the set of control variables; γ_j represents industry-fixed effects to control for time-invariable unobservable factors at the industry level; λ_t represents

time-fixed effects to control for unobservable factors that change over time; ε_{ijt} is the random error term.

Environmental regulation is a good way to correct corporate environmental behavior, prompting companies to actively invest in environmental initiatives. However, due to the market-driven nature of corporate social responsibility, environmental regulatory pressure may affect the execution of corporate environmental responsibility in various ways. First, in the context of sustainable development, companies must effectively address ESG issues to stand out in an increasingly fierce global market. The integration of sustainable development and responsible investment concepts with environmental protection has seen the gradual emergence of ESG. Second, companies may respond to the pressure of environmental regulation by disguising themselves as "friends of the environment" to cover up any social and environmental damage they have inflicted, and maintaining and expanding their market or influence as a result. Therefore, to examine the impact of environmental regulation on corporate green innovation, we employ corporate ESG ratings and greenwashing behavior scores as mediating variables and construct the following mediation effect models:

$$M_{it} = \beta_0 + \beta_1 EEFI_{ijt} + \beta_2 Control_{it} + \gamma_j + \lambda_t + \varepsilon_{ijt} \quad (2)$$

$$GreenPatent_{ijt} = \zeta_0 + \zeta_1 EEFI_{ijt} + \zeta_2 M_{it} + \zeta_3 Control_{ijst} + \gamma_j + \lambda_t + \varepsilon_{ijt}, \quad (3)$$

where M_{it} is the mediating variable (corporate ESG ratings and greenwashing behavior scores); β_1 measures the impact of environmental regulation on the development of green finance, corporate ESG ratings, and greenwashing behavior. Table 2 provides descriptive statistics for our variables.

4. Empirical results and discussion

4.1. Unit root test

We conduct Fisher unit root tests on the variables using both the dfuller and pperon methods to avoid the phenomenon of spurious regression. Table 3 presents the panel unit root test results for the dependent, independent, control, and mediating variables. The results indicate a *p*-value of 0.0000 for the Fisher unit root tests of all variables, including the dependent variable (corporate green innovation, *GreenPatent*), the independent variable (environmental regulation, *EEFI*), and control variables of firm size (*Size*), ownership type (*Soe*), cash flow (*Cflow*), equity concentration (*Herf*), asset structure (*Stru*), return on assets (*Roa*), growth ability (*Growth*), and Tobin's Q (*TQ*), as well as the mediating variables greenwashing behavior (*GreenWash*) and corporate ESG performance (*ESG*). This finding indicates that these variables are stationary in the panel data, providing a solid foundation for conducting regression analysis to explore the influence of environmental regulation on corporate green innovation.

Table 2
Descriptive Statistics of Main Variables.

Variable	Obs	Mean	Std. Dev.	Min	Max
GreenPatent	20,489	0.367	0.788	0.000	4.007
EEFI	20,489	0.136	0.121	0.000	0.922
Size	20,489	3.097	0.058	2.957	3.260
Soe	20,489	0.131	0.337	0.000	1.000
Cflow	20,489	0.046	0.071	−0.192	0.253
Herf	20,489	3.452	0.466	2.172	4.313
Stru	20,489	0.371	0.182	0.016	0.820
Roa	20,489	0.042	0.060	−0.239	0.228
Growth	20,489	0.198	0.522	−0.659	3.807
TQ	20,489	0.597	0.485	−0.144	2.167
GreenWash	20,489	0.451	0.277	0.000	1.253
ESG	20,489	4.089	0.991	1.000	7.750

Note: Compiled by the author.

Table 3
Unit Root Test Results.

Variable	Fisher dfuller		Fisher pperon	
	Pm	P	Pm	P
GreenPatent	61.0750	0.0000	73.6893	0.0000
EEFI	58.9496	0.0000	54.9193	0.0000
Size	51.2138	0.0000	67.1336	0.0000
Soe	18.1317	0.0000	102.2702	0.0000
Cflow	66.8142	0.0000	119.4668	0.0000
Herf	54.4401	0.0000	44.3063	0.0000
Stru	59.9979	0.0000	77.8081	0.0000
Roa	61.0988	0.0000	87.9020	0.0000
Growth	72.2772	0.0000	125.8899	0.0000
TQ	60.9641	0.0000	49.1691	0.0000
GreenWash	64.1630	0.0000	143.1491	0.0000
ESG	59.0081	0.0000	51.7606	0.0000

Note: Compiled by the author.

Table 4
Baseline Regression Results.

Variable	(1)	(2)	(3)	(4)
	GreenPatent	*GreenPatent*	*GreenPatent*	*GreenPatent*
EEFI	−0.1182***	−0.1243***	−0.0840**	−0.0847**
	(0.0455)	(0.0442)	(0.0297)	(0.0278)
Size		3.6056***		4.3308***
		(0.1130)		(0.2137)
Soe		−0.0489***		0.0065
		(0.0162)		(0.0121)
Cflow		0.0065		0.0250
		(0.0812)		(0.0465)
Herf		−0.0210*		−0.0073
		(0.0119)		(0.0139)
Stru		−0.4585***		−0.0967***
		(0.0308)		(0.0263)
Roa		0.1569		0.3812***
		(0.1011)		(0.1025)
Growth		−0.0577***		−0.0591***
		(0.0104)		(0.0098)
TQ		0.0525***		0.0985***
		(0.0136)		(0.0156)
Constant	0.3833***	−10.5604***	0.3786***	−13.0386***
	(0.0083)	(0.3511)	(0.0040)	(0.6719)
Year fixed effects	N	N	Y	Y
Industry fixed effects	N	N	Y	Y
N	20,489	20,489	20,487	20,487
adj. R^2	0.000	0.064	0.145	0.211

Note:* , ** , *** represent significance levels of 10 %, 5 %, and 1 %, respectively; robust standard errors are reported in parentheses.

4.2. Baseline regression results

Table 4 presents the results for the regression examining the impact of environmental regulation on corporate green innovation. The regression results reveal that the coefficient estimates for *EEFI* are all significant, at least at the 5 % level, indicating that environmental regulation significantly enhances corporate green innovation. Specifically, after controlling for year and industry fixed effects, environmental regulation increases corporate green innovation by 0.0840 and 0.0847 units when control variables are not included and when they are, respectively.

Overall, environmental regulation has a significant positive impact on corporate green innovation, confirming Hypothesis 1. This impact can be attributed to several factors. First, environmental regulation increases firms' production costs. For instance, the government may require firms to reduce pollutant emissions, improve waste treatment efficiency, or adopt more environmentally friendly production technologies and equipment, all of which increase production costs. Firms are then compelled to invest in green technologies and innovation to

Table 5
Endogeneity Test Results.

Variable	(1)	(2)	(3)	(4)
	First	Second	First	Second
	EEFI	*GreenPatent*	*EEFI*	*GreenPatent*
EEFI		−0.1204*		−0.1107*
		(0.0623)		(0.0602)
EEFI_after	0.8668***		0.8666***	
	(0.0070)		(0.0071)	
Size			−0.0062	3.9972***
			(0.0093)	(0.1825)
Soe			−0.0012	−0.0214
			(0.0015)	(0.0211)
Cflow			0.0089	0.0221
			(0.0067)	(0.0894)
Herf			0.0007	−0.0275*
			(0.0009)	(0.0146)
Stru			0.0041	−0.4108***
			(0.0026)	(0.0348)
Roa			0.0066	0.1093
			(0.0080)	(0.1008)
Growth			0.0007	−0.0625***
			(0.0009)	(0.0085)
TQ			−0.0008	0.1005***
			(0.0012)	(0.0168)
Constant	0.0272***	0.1794***	0.0423	−11.8993***
	(0.0028)	(0.0307)	(0.0288)	(0.5645)
Kleibergen-Paap rk LM statistic	2546.313 [0.0000]		2539.724 [0.0000]	
Kleibergen-Paap rk Wald F statistic	15,000		15,000	
Observations	15,808	15,808	15,808	15,808
R-squared		0.006		0.067

Note: *, **, *** represent significance levels of 10 %, 5 %, and 1 %, respectively; robust standard errors are reported in () and *p*-values in [].

Table 6
Replacement of Environmental Regulation Measurement Variables.

Variable	(1)	(2)	(3)	(4)
	GreenPatent	*GreenPatent*	*GreenPatent*	*GreenPatent*
Ser	0.2090***	0.1437***	0.1161***	0.0997***
	(0.0199)	(0.0198)	(0.0235)	(0.0259)
Size		3.5052***		4.5173***
		(0.1243)		(0.1728)
Soe		−0.0380**		0.0137
		(0.0188)		(0.0129)
Cflow		0.0074		0.0058
		(0.0912)		(0.0517)
Herf		−0.0126		0.0035
		(0.0130)		(0.0154)
Stru		−0.4294***		−0.0774**
		(0.0345)		(0.0331)
Roa		0.2148*		0.4008***
		(0.1115)		(0.0864)
Growth		−0.0621***		−0.0697***
		(0.0118)		(0.0118)
TQ		0.0381**		0.1010***
		(0.0148)		(0.0167)
Constant	−0.6366***	−10.9935***	−0.1811	−14.1464***
	(0.0977)	(0.3867)	(0.1154)	(0.6027)
Year fixed effects	N	N	Y	Y
Industry fixed effects	N	N	Y	Y
N	18,105	18,105	18,101	18,101
adj. R^2	0.006	0.062	0.144	0.212

Note:* , ** , *** represent significance levels of 10 %, 5 %, and 1 %, respectively; robust standard errors are reported in parentheses.

lower costs and comply with regulations. Environmental regulation pressures firms to compensate for potential negative impacts by enhancing their R&D efficiency and innovating in environmental management technologies, thereby improving the eco-friendliness of their production processes to meet energy-saving and emission-reduction targets (Qi et al., 2018; Deng et al., 2021). Second, environmental regulation alters market demand. As consumers have become more concerned about environmental protection and sustainable development, their demand for environmentally friendly products and services has increased. The implementation of environmental regulations makes

Table 7
Replacement of Regression Model Methods.

Variable	PPML		MLE	
	(1)	(2)	(3)	(4)
	GreenPatent	*GreenPatent*	*GreenPatent*	*GreenPatent*
EEFI	−0.3351***	−0.3082**	−0.0840*	−0.0847*
	(0.1265)	(0.1230)	(0.0457)	(0.0439)
Size		8.7561***		4.3308***
		(0.2848)		(0.1185)
Soe		−0.1204***		0.0065
		(0.0435)		(0.0154)
Cflow		0.0519		0.0250
		(0.2145)		(0.0775)
Herf		−0.0770**		−0.0073
		(0.0320)		(0.0112)
Stru		−1.2750***		−0.0967***
		(0.0777)		(0.0337)
Roa		0.7409***		0.3812***
		(0.2666)		(0.0949)
Growth		−0.1777***		−0.0591***
		(0.0271)		(0.0097)
TQ		0.0707*		0.0985***
		(0.0368)		(0.0149)
Constant	−0.9571***	−27.4992***	−0.0274	−13.3184***
	(0.0225)	(0.8886)	(0.0805)	(0.3767)
N	20,489	20,489	20,489	20,489
R^2	0.000	0.075		
Log likelihood			−22,534.087	−21,708.694
Chi - Square Test of Model Fit			3583.16 [0.0000]	5602.91 [0.0000]

Note: *, **, *** represent significance levels of 10 %, 5 %, and 1 %, respectively; robust standard errors are reported in () and p-values in [].

Table 8
Sample Period Adjustment (2012–2019).

Variable	(1)	(2)	(3)	(4)
	GreenPatent	*GreenPatent*	*GreenPatent*	*GreenPatent*
EEFI	−0.0600	−0.1140**	−0.0976**	−0.1011***
	(0.0519)	(0.0505)	(0.0293)	(0.0284)
Size		3.6011***		4.4893***
		(0.1236)		(0.1915)
Soe		−0.0406**		0.0134
		(0.0187)		(0.0146)
Cflow		−0.0076		0.0051
		(0.0914)		(0.0522)
Herf		−0.0212		−0.0033
		(0.0129)		(0.0158)
Stru		−0.4417***		−0.0770**
		(0.0341)		(0.0264)
Roa		0.2510**		0.4756***
		(0.1116)		(0.0875)
Growth		−0.0586***		−0.0672***
		(0.0117)		(0.0106)
TQ		0.0493***		0.0994***
		(0.0147)		(0.0166)
Constant	0.3944***	−10.5481***	0.3994***	−13.5405***
	(0.0091)	(0.3836)	(0.0039)	(0.5982)
Year fixed effects	N	N	Y	Y
Industry fixed effects	N	N	Y	Y
N	18,277	18,277	18,276	18,276
adj. R^2	0.000	0.059	0.143	0.211

Note:* , ** , *** represent significance levels of 10 %, 5 %, and 1 %, respectively; robust standard errors are reported in parentheses.

Table 9
Mediating Effect of Greenwashing Behavior on Environmental Regulation and Green Innovation.

Variable	(1)	(2)	(3)	(4)	(5)	(6)
	GreenPatent	*GreenWash*	*GreenPatent*	*GreenPatent*	*GreenWash*	*GreenPatent*
EEFI	−0.0840**	−0.0309	−0.0749**	−0.0847**	−0.0281	−0.0829**
	(0.0297)	(0.0227)	(0.0307)	(0.0278)	(0.0241)	(0.0279)
GreenWash			0.2967***			0.0652***
			(0.0271)			(0.0164)
Size				4.3308***	1.7051***	4.2196***
				(0.2137)	(0.0814)	(0.2021)
Soe				0.0065	0.0124	0.0057
				(0.0121)	(0.0099)	(0.0120)
Cflow				0.0250	0.0217	0.0236
				(0.0465)	(0.0303)	(0.0458)
Herf				−0.0073	−0.0000	−0.0072
				(0.0139)	(0.0030)	(0.0140)
Stru				−0.0967***	0.0769***	−0.1017***
				(0.0263)	(0.0091)	(0.0259)
Roa				0.3812***	0.0498	0.3780***
				(0.1025)	(0.0321)	(0.1024)
Growth				−0.0591***	−0.0283***	−0.0572***
				(0.0098)	(0.0066)	(0.0096)
TQ				0.0985***	0.0303***	0.0965***
				(0.0156)	(0.0092)	(0.0155)
Constant	0.3786***	0.4549***	0.2437***	−13.0386***	−4.8722***	−12.7207***
	(0.0040)	(0.0031)	(0.0107)	(0.6719)	(0.2543)	(0.6403)
Year fixed effects	Y	Y	Y	Y	Y	Y
Industry fixed effects	Y	Y	Y	Y	Y	Y
N	20,487	20,487	20,487	20,487	20,487	20,487
adj. R^2	0.145	0.105	0.155	0.211	0.196	0.211

Note:* , ** , *** represent significance levels of 10 %, 5 %, and 1 %, respectively; robust standard errors are reported in parentheses.

Table 10
Mediating Effect of Corporate ESG on Environmental Regulation and Green Innovation.

Variable	(1)	(2)	(3)	(4)	(5)	(6)
	GreenPatent	*ESG*	*GreenPatent*	*GreenPatent*	*ESG*	*GreenPatent*
EEFI	−0.0840**	−0.1290***	−0.0672**	−0.0847**	−0.1689***	−0.0714**
	(0.0297)	(0.0370)	(0.0281)	(0.0278)	(0.0446)	(0.0271)
ESG			0.1309***			0.0788***
			(0.0090)			(0.0096)
Size				4.3308***	4.2586***	3.9950***
				(0.2137)	(0.2046)	(0.2032)
Soe				0.0065	−0.0200	0.0081
				(0.0121)	(0.0331)	(0.0116)
Cflow				0.0250	0.3044***	0.0010
				(0.0465)	(0.0726)	(0.0447)
Herf				−0.0073	0.0745***	−0.0131
				(0.0139)	(0.0154)	(0.0127)
Stru				−0.0967***	0.0112	−0.0976***
				(0.0263)	(0.0359)	(0.0264)
Roa				0.3812***	3.2237***	0.1271
				(0.1025)	(0.2210)	(0.0879)
Growth				−0.0591***	−0.1745***	−0.0453***
				(0.0098)	(0.0193)	(0.0082)
TQ				0.0985***	−0.1432***	0.1098***
				(0.0156)	(0.0419)	(0.0167)
Constant	0.3786***	4.1062***	−0.1588***	−13.0386***	−9.3652***	−12.3002***
	(0.0040)	(0.0050)	(0.0357)	(0.6719)	(0.6469)	(0.6471)
Year fixed effects	Y	Y	Y	Y	Y	Y
Industry fixed effects	Y	Y	Y	Y	Y	Y
N	20,487	20,487	20,487	20,487	20,487	20,487
adj. R^2	0.145	0.078	0.170	0.211	0.187	0.219

Note:* , ** , *** represent significance levels of 10 %, 5 %, and 1 %, respectively; robust standard errors are reported in parentheses.

green products and services mainstream choices in the market. Firms will increase their investment in green innovation to develop more eco-friendly products and services to meet market demand and gain a competitive advantage(Zhang et al., 2024). Finally, firms that have adopted environmental technologies and products often have a competitive edge. Being seen as eco-friendly helps firms build a positive brand image and attract more consumers and investors. This competitive advantage further incentivizes firms to increase their investment in green innovation.

4.3. Robustness checks

4.3.1. Instrumental variable method

To mitigate estimation bias arising from bidirectional causality and omitted variable issues, we apply the instrumental variable method in our robustness checks. Drawing on the methods of Du et al. (2019) and

Table 11
Different Levels of Government Intervention.

Variable	High Government Intervention		Low Government Intervention	
	(1)	(2)	(3)	(4)
	GreenPatent	*GreenPatent*	*GreenPatent*	*GreenPatent*
EEFI	0.0028	0.0209	−0.2680***	−0.2749***
	(0.0473)	(0.0488)	(0.0406)	(0.0429)
Size		4.2007***		4.5903***
		(0.2118)		(0.2943)
Soe		−0.0172		0.0320
		(0.0239)		(0.0223)
Cflow		−0.0569		0.0281
		(0.0843)		(0.0512)
Herf		−0.0228		−0.0025
		(0.0158)		(0.0209)
Stru		−0.0349		−0.2061***
		(0.0317)		(0.0311)
Roa		0.4275***		0.3685**
		(0.1348)		(0.1199)
Growth		−0.0588***		−0.0588***
		(0.0142)		(0.0114)
TQ		0.0882***		0.1191***
		(0.0159)		(0.0208)
Constant	0.3728***	−12.6277***	0.3997***	−13.7791***
	(0.0060)	(0.6504)	(0.0057)	(0.9297)
Year fixed effects	Y	Y	Y	Y
Industry fixed effects	Y	Y	Y	Y
N	9123	9123	10,597	10,597
adj. R^2	0.158	0.222	0.147	0.215

Note:* , ** , *** represent significance levels of 10 %, 5 %, and 1 %, respectively; robust standard errors are reported in parentheses.

Shi and Yi (2020), we use the lagged value of environmental regulation as an instrumental variable for the current value, employing two-stage least squares regression to address potential endogeneity concerns. Theoretically, using the lagged value as an instrumental variable is appropriate because: (i) the lagged and current values of environmental regulation are highly correlated, and (ii) the lagged value is exogenous to the current level of corporate green innovation, satisfying the validity criteria for instrumental variables.

Table 5 presents the results of the endogeneity tests. TThe Kleibergen-Paap rk LM statistic and Kleibergen-Paap rk Wald F statistic in the first stage confirm that the instrumental variable is both identifiable and free from weak instrument issues. The second-stage results indicate that, whether or not control variables are included, environmental regulation significantly promotes corporate green innovation after accounting for endogeneity. This finding aligns with the baseline regression results, underscoring the robustness of our conclusions.

4.3.2. *Replacing environmental regulation measurement variables*

In the preceding section, we quantified environmental regulation using pollutant emission indicators at the prefecture level and employed the entropy method to calculate a comprehensive environmental regulation index. While this indicator is broad and comprehensive, the result is significantly impacted by the method of calculation and variable indicators. We ensure the robustness of empirical tests by replacing the basic indicator with a socially driven informal environmental regulation indicator, using the Baidu Index of Environmental Pollution (*Ser*) as a proxy for environmental regulation. The Baidu Index, based on netizen behavior data, objectively reflects social trends and hotspots. Following the method of Xu (2014), we select “environmental pollution” as a keyword and retrieve daily data for 34 provinces from 2011 to 2019 from the Baidu Index website, calculating the annual average and matching it with corporate addresses and province codes. A higher *Ser* indicates higher levels of public attention and more efficient environmental regulation, and vice versa. Table 6 presents the regression results after substituting the proxy variable for environmental regulation. The coefficient estimates for Ser in Columns (1) and (2) are both positive and significant at the 1 % level. Similarly, in Columns (3) and (4), the estimates remain significantly positive, indicating that environmental regulation consistently exerts a significant positive effect on corporate green innovation, irrespective of the inclusion of control variables or fixed effects. These findings reaffirm that the influence of environmental regulation on corporate green innovation at the provincial level aligns with our main conclusions.

Table 12
Corporate Ownership.

Variable	State-owned		Private		Foreign-owned		Others	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	GreenPatent	*GreenPatent*	*GreenPatent*	*GreenPatent*	*GreenPatent*	*GreenPatent*	*GreenPatent*	*GreenPatent*
EEFI	0.1797***	0.1339**	−0.2198***	−0.2166***	−0.0056	−0.0203	0.5630	0.4276
	(0.0561)	(0.0549)	(0.0324)	(0.0393)	(0.1135)	(0.1020)	(0.3414)	(0.3433)
Size		4.8912***		3.5106***		2.1307***		5.7932***
		(0.3179)		(0.2334)		(0.5519)		(1.0101)
Soe		−0.0220		0.0169		0.0000		−0.1985
		(0.0131)		(0.1329)		(.)		(0.1560)
Cflow		0.1974		0.0072		−0.6599***		−0.6091
		(0.1404)		(0.0605)		(0.1879)		(0.6902)
Herf		−0.0419**		−0.0154		0.1072**		0.1280**
		(0.0186)		(0.0176)		(0.0429)		(0.0517)
Stru		−0.0781***		−0.0767*		−0.2903		−0.1838
		(0.0182)		(0.0356)		(0.1743)		(0.1898)
Roa		0.3216		0.5950***		0.1076		0.6287
		(0.2345)		(0.1179)		(0.4657)		(0.3886)
Growth		−0.0596***		−0.0538***		0.0175		−0.0856**
		(0.0167)		(0.0085)		(0.0436)		(0.0339)
TQ		0.1142***		0.0769***		0.1467*		0.1043*
		(0.0279)		(0.0143)		(0.0715)		(0.0484)
Constant	0.3767***	−14.7633***	0.3756***	−10.4276***	0.2848***	−6.6218***	0.2818***	−18.0516***
	(0.0075)	(1.0185)	(0.0044)	(0.7216)	(0.0192)	(1.6513)	(0.0511)	(3.2092)
Year fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
Industry fixed effects	Y	Y	Y	Y	Y	Y	Y	Y
N	8102	8102	11,041	11,041	658	658	601	601
adj. R^2	0.202	0.277	0.155	0.194	0.209	0.225	0.308	0.383

Note:* , ** , *** represent significance levels of 10 %, 5 %, and 1 %, respectively; robust standard errors are reported in parentheses.

Table 13
Whether the Firm Belongs to the Energy Industry.

Variable	Non-Energy Firms		Energy Firms		New Energy Firms	
	(1)	(2)	(3)	(4)	(5)	(6)
	GreenPatent	*GreenPatent*	*GreenPatent*	*GreenPatent*	*GreenPatent*	*GreenPatent*
EEFI	0.0539	0.0279	−0.3285***	−0.2358**	−1.2383***	−1.4421***
	(0.0390)	(0.0394)	(0.0932)	(0.0946)	(0.3310)	(0.2607)
Size		3.1712***		6.4453***		6.7167***
		(0.1522)		(0.4479)		(0.7912)
Soe		0.0223		−0.0305		−0.0960
		(0.0172)		(0.0292)		(0.0638)
Cflow		−0.1375**		0.2944**		1.3180
		(0.0518)		(0.1151)		(0.9977)
Herf		−0.0229		0.0143		0.0620
		(0.0127)		(0.0275)		(0.0775)
Stru		−0.1129***		−0.1600		0.3426**
		(0.0299)		(0.0898)		(0.1280)
Roa		0.4192***		0.1266		1.7202***
		(0.0831)		(0.2036)		(0.5396)
Growth		−0.0474***		−0.0673***		−0.1966***
		(0.0094)		(0.0162)		(0.0556)
TQ		0.0332*		0.2970***		0.0461
		(0.0163)		(0.0375)		(0.1277)
Constant	0.2909***	−9.4241***	0.5315***	−19.6155***	0.9600***	−20.4514***
	(0.0053)	(0.4699)	(0.0128)	(1.4399)	(0.0405)	(2.5908)
Year fixed effects	Y	Y	Y	Y	Y	Y
Industry fixed effects	Y	Y	Y	Y	Y	Y
N	14,483	14,483	5119	5119	880	880
adj. R^2	0.158	0.204	0.146	0.248	0.150	0.233

Note:* , ** , *** represent significance levels of 10 %, 5 %, and 1 %, respectively; robust standard errors are reported in parentheses.

Table 1
Variable Definitions.

Variable Type	Variable Name	Variable Definition
Dependent Variable	*GreenPatent*	ln(Green Patent Applications +1)
Core Independent Variable	*EEFI*	Comprehensive Environmental Regulation Index, calculated using the entropy method based on city-level unit pollutant emissions indicators, including industrial wastewater per unit output, SO_2 emissions per unit output, and industrial soot and dust emissions per unit output.
Mechanism Variable	*ESG*	Huazheng ESG quarterly rating, ranging from C to AAA, converted to values from 1 to 9, with annual average/median used.
	GreenWash	Based on the research by Walker and Fang (2012) and Huang et al. (2019, 2020), measured as the ratio of symbolic to substantive actions in corporate environmental governance, based on dimensions of governance and structure, processes and controls, inputs and outputs, and compliance (see Appendix B for detailed calculation).
Control Variable	*Size*	Natural logarithm of total assets
	Soe	Categorized by ownership type: state-owned or non-state-owned
	Cflow	Ratio of cash flow from operating activities to total assets
	Herf	Shareholding ratio of the largest shareholder
	Stru	Ratio of net fixed assets plus net inventory to total assets
	Roa	Ratio of net profit to average total assets, where average total assets is the mean of the beginning and ending total assets
	Growth	Annual average of the quarterly revenue growth rate
	TQ	Ratio of market value to total assets

4.3.3. Replacing regression methods

In fixed-effects models, heteroscedasticity may lead to inconsistent and inefficient estimation results. The Poisson pseudo-maximum likelihood estimation (PPML) naturally handles heteroscedasticity in the model. Compared to fixed-effects models, PPML does not require heteroscedasticity-adjusted standard errors and can provide more robust estimation results. Additionally, under certain conditions, maximum likelihood estimation (MLE) is consistent and asymptotically efficient, meaning that as the sample size increases, the estimator converges to the true parameter; it is most efficient in large samples. The flexibility of the MLE method also allows it to be adapted to different distribution assumptions and complex model structures. Therefore, to ensure that our conclusions hold under different model assumptions and to improve the reliability and credibility of the results, we employ PPML and MLE for robustness checks. In Table 7, columns (1) and (2) present the PPML results, and Columns (3) and (4) show the MLE results. The results indicate that, regardless of whether PPML or MLE estimation is used, the results for the regression of *EEFI* on *GreenPatent* are significant, with negative coefficients; this is consistent with the baseline regression results, confirming their robustness.

4.3.4. Sample adjustment

Environmental issues have received unprecedented attention from China's Communist Party and government since the Party's 18th National Congress in 2012, and the country's environmental protection efforts have since reached new heights. The 18th National Congress proposed the major task of constructing an ecological civilization and intensified efforts to promote environmental protection and green development at the policy level. Since 2012, China has implemented increasingly stringent environmental regulations and policies. To test the robustness of the previous regression results, we adjust the sample period and conduct the analysis using data from 2012 to 2019. The results are presented in Table 8. The regression results are structured into four columns: odd-numbered columns exclude control variables, while even-numbered columns include them. Columns (1) and (2) show the OLS regression results, and Columns (3) and (4) account for year- and industry-fixed effects. The results in Table 8 indicate that, after

controlling for fixed effects and including control variables, the coefficient estimates for *EEFI* remain negative and significant, at least at the 5 % level. This suggests that, even with the adjusted sample period, environmental regulation continues to significantly promote corporate green innovation. Therefore, the baseline regression results remain robust regardless of the time range of the sample.

5. Mechanism tests and heterogeneity discussion

5.1. Mediating effects test results

Greenwashing, as a speculative behavior where corporations present a false image of social responsibility, is influenced by environmental regulations and can also affect corporate green innovation. This section investigates the mediating role of greenwashing in the relationship between environmental regulation and green innovation.

Table 9 displays the regression results assessing the mediating effect of greenwashing on the link between environmental regulation and corporate green innovation. Control variables are excluded in Columns (1) to (3) and included in Columns (4) to (6). First, in Columns (1) and (4), the coefficient estimates for *EEFI* are significant at the 5 % level, indicating that environmental regulation positively influences corporate green innovation. Second, in Columns (3) and (6), the coefficient estimates for *GreenWash* are positive and significant, while those for *EEFI* turn negative and significant. Although the coefficients for *GreenWash* in Columns (2) and (5) are negative but not significant, we conducted a bootstrap test, following the methodology of Wen and Ye (2014), and the results confirm that the mediating effect is indeed significant. These findings suggest that greenwashing exhibits a partial but significant mediating effect in profit-driven green innovation activities; that is, environmental regulations can stimulate corporate greenwashing behavior, which in turn promotes green innovation, thus confirming Hypothesis 2. This might be because strengthened environmental regulations impose greater compliance pressure on companies. Companies might engage in greenwashing to demonstrate their environmental commitments in the interest of maintaining their public image and reputation. Despite the speculative nature of greenwashing, companies often invest resources in actual green innovation to make greenwashing more credible. This limited investment can have some positive effects.

Corporate ESG performance serves as a key measure of social responsibility and environmental commitment. We further analyze the impact of environmental regulations on corporate green innovation by exploring the mediating role of ESG performance.

Table 10 presents the regression results examining the mediating effect of ESG performance on the relationship between environmental regulation and corporate green innovation. Columns (1) to (3) exclude control variables, while Columns (4) to (6) include a range of control variables. First, in Columns (1) and (4), the coefficient estimates for *EEFI* are significant at the 5 % level, suggesting that environmental regulation positively influences corporate green innovation. Second, in Columns (2) and (5), the coefficient estimates for *EEFI* are negative and significant, indicating that environmental regulation significantly enhances ESG performance. Finally, in Columns (3) and (6), the coefficient estimates for ESG are positive and significant, while those for *EEFI* become significantly negative. The mediating effect analysis indicates that environmental regulation promotes corporate green innovation by improving ESG performance, thereby validating Hypothesis 3. This outcome may be due to stricter environmental regulations prompting companies to prioritize their ESG performance. Enhanced efforts in environmental practices, social responsibility, and governance not only elevate a company's reputation but also boost competitiveness. Consequently, companies are driven to pursue green innovation to comply with regulatory standards and align with societal expectations. Moreover, achieving strong ESG performance often necessitates resource reallocation and optimization, including investments in R&D and technological innovation, leading to meaningful advancements in green innovation.

5.2. Heterogeneity analysis

Given that environmental regulation entails the development and enforcement of relevant policies and frameworks, its impact on companies can differ based on their economic environment and specific characteristics. Therefore, we further investigate the heterogeneous effects of environmental regulation on corporate green innovation by examining variations according to the level of government intervention, ownership structure, and whether the company operates within the energy sector.

5.2.1. Government intervention levels

The level of government intervention indicates the extent of government involvement in market activities. Following the approach of Guo et al. (2020), this study measures government intervention using the ratio of local fiscal expenditure to regional GDP. Based on the median value of government intervention, the sample is divided into high-intervention and low-intervention groups. The results are presented in Table 11.

The results show that in high-intervention companies, the coefficients for *EEFI* are not significant, while in low-intervention companies, the coefficients for *EEFI* are significant and negative. These results indicate that when government intervention is high, environmental regulations do not significantly affect corporate green innovation. In contrast, when government intervention is low, environmental regulations significantly promote green innovation. The possible reason for this is that when the level of government intervention in the economy is high, companies may rely on policy support, subsidies, and other forms of support rather than independently engaging in green innovation. In such cases, environmental regulations may be seen as compliance requirements rather than incentives for innovation. Green innovation also requires a favorable external environment, and excessive government intervention can lead to market and subsidy distortions, reducing labor and capital returns and hindering R&D and innovation. Conversely, in a low-intervention environment, market mechanisms play a larger role(Zhang et al., 2024). Environmental regulations convey that companies need to reduce pollution and increase efficiency to adapt to market demands and regulatory pressures, and companies will then proactively engage in green innovation to gain a competitive advantage.

5.2.2. Corporate ownership nature

This study categorizes the sample by ownership type into state-owned, private, foreign, and other enterprises. The estimation results are presented in Table 12. Columns (1) and (2) show positive and significant coefficients for *EEFI*, suggesting that environmental regulations significantly hinder green innovation in state-owned enterprises. Conversely, Columns (3) and (4) display negative and significant coefficients for *EEFI*, indicating that environmental regulations significantly promote green innovation in private enterprises. Columns (5) to (7) reveal non-significant coefficients for *EEFI*, indicating that environmental regulations do not significantly impact green innovation in foreign and other enterprises. A comparison of these results highlights that environmental regulations have a significant inhibitory effect on green innovation in state-owned enterprises, while they substantially encourage green innovation in private enterprises, with minimal impact on foreign and other enterprises.

In state-owned enterprises, the significant inhibitory effect of environmental regulations on green innovation might be due to the firms' reliance on government policy support and subsidies. When environmental regulations are strengthened, state-owned enterprises may see these as compliance burdens rather than incentives to innovate, with excessive intervention potentially leading to distorted resource allocation and inhibited innovation capacity. In contrast, with enhanced

environmental regulation, private enterprises face stricter external environment constraints. To meet environmental standards and maintain long-term operations, companies will engage in technological innovation to reduce negative externalities in production. Foreign enterprises, which have global operational experience and resources, can flexibly respond to changes in environmental regulations, minimizing disruption to green innovation. Other enterprises might already have high standards of environmental management and advanced technology, with regulations having a limited impact on green innovation.

5.2.3. Whether the company is in the energy sector

We examine the heterogeneity in green innovation responses to environmental regulation across different types of companies by categorizing the sample into non-energy, energy, and new energy companies. This classification allows for a more precise analysis of how environmental regulations affect various types of companies and the underlying mechanisms. Table 13 presents the corresponding results. Columns (1) and (2) show the regression results for non-energy companies, Columns (3) and (4) show the results for energy companies, and Columns (5) and (6) display the results for new energy companies. Odd-numbered columns exclude control variables, while even-numbered columns include them.

In the non-energy company sample, the coefficient estimate for *EEFI* is 0.0539 and is not statistically significant. After including control variables, the coefficient changes to 0.0279 but remains insignificant, indicating that environmental regulations do not significantly impact green innovation in non-energy companies. For energy companies, the estimated coefficient for *EEFI* is −0.3285 and is significantly negative at the 1 % level. When control variables are added, the coefficient adjusts to −0.2358 and remains negative and significant at the 5 % level, suggesting that environmental regulations have a notable positive effect on green innovation in energy companies. In the new energy company sample, the estimated coefficient for *EEFI* is −1.2383, which is significant at the 1 % level. After accounting for control variables, the coefficient changes to −1.4421 and remains significantly negative at the 1 % level. This outcome indicates that environmental regulations exert a substantial positive effect on green innovation in new energy companies, and this positive impact is more pronounced compared to that in energy companies.

In non-energy companies, the insignificant impact of environmental regulation on green innovation might be due to these companies having less environmental impact or having already adopted high environmental standards, limiting the incentive effect of regulations. In energy companies, the significant and positive impact of environmental regulation on green innovation is due to stricter regulations compelling these companies to innovate to reduce pollution and improve efficiency. In new energy companies, the more significant positive impact of environmental regulations on green innovation might be because these companies focus on green innovation as a core competitive advantage, and stricter regulations further promote their technological development and innovation. Market demand and policy support also establish favorable conditions for green innovation by new energy companies, where high levels of intervention promote innovation.

6. Conclusion and outlook

China's economy has gradually shifted from being quantity and speed-driven to one focused on quality, efficiency, and sustainability. Balancing environmental protection with the establishment of a green economy has become a crucial area of inquiry in this context. Building on this, we used a sample of listed companies from 2007 to 2019 to explore the relationship between environmental regulation and green innovation in Chinese enterprises. Our research objectives were as follows: First, we detailed the development of environmental regulations in China and examined the characteristics and background of green innovation among Chinese enterprises. Second, we established the theoretical framework of the study through a comprehensive review of relevant theories and literature on the impact of environmental regulations on corporate innovation. Third, we constructed an index system to evaluate environmental regulations and corporate green innovation based on real-world scenarios and theoretical insights. Finally, we conducted an empirical analysis to assess the impact of environmental regulations on green innovation in Chinese enterprises, further investigating how this impact is mediated by corporate greenwashing and ESG performance.

The main conclusions of this study are as follows: First, environmental regulations positively influence green innovation in Chinese enterprises. We tested the robustness of our baseline results using instrumental variables, alternative proxy indicators for core explanatory variables, different model estimation methods, and adjustments to the research sample. The consistent results across these tests confirm the reliability of our findings. Second, corporate greenwashing and ESG performance play mediating roles in the relationship between environmental regulation and green innovation. Greenwashing can reduce companies' regulatory burden, encouraging green innovation while alleviating reputational pressures imposed by environmental regulations. Additionally, enterprises facing regulatory pressure that actively fulfill their ESG responsibilities are more likely to engage in green innovation. Third, the heterogeneity analysis reveals that the impact of environmental regulations on green innovation varies across different economic environments, corporate characteristics, and industry attributes. Specifically, the positive effect of environmental regulations on green innovation is stronger in regions with lower levels of government intervention. Moreover, environmental regulations have a more pronounced effect on promoting green innovation in private enterprises compared to state-owned and foreign enterprises and show a greater impact on green innovation in energy and new energy enterprises than in non-energy enterprises.

Our findings have several policy implications, and we offer several recommendations based on our conclusions. First, we consider the perspective of environmental regulation policies. Environmental regulations offer one pathway to achieving the dual carbon goals of “carbon peak by 2030 and carbon neutrality by 2060.” Our findings show that environmental regulations are beneficial for corporate green innovation. However, theoretical analysis reveals that the intensity of environmental regulations must be within a reasonable range to effectively promote green innovation in enterprises. Policies should thus avoid a “one-size-fits-all” approach and instead create tailored and supportive legal environments. The government should establish and improve systems of environmental evaluation and assessment standards within its jurisdiction, focusing on the adaptability of incentive mechanisms and constraint schemes. Further, the government should emphasize building an environment that is conducive to innovation. While ensuring environmental compliance, it is also important to protect and foster the entrepreneurial spirit. Furthermore, policies should be established that are aligned with low-carbon emission-reduction and circular development goals, including supporting efficient systems of dispute resolution and emergency response to provide legal and institutional guarantees for green production and quality transformation by enterprises.

Second, our findings have implications concerning the drivers of green innovation. With ongoing economic development, technological innovation significantly boosts a country's high-quality development. The government should increase investment and support for green technology R&D, including by establishing special funds and research projects to encourage universities, research institutions, and enterprises to jointly conduct green technology research and development. By providing research funding, tax incentives, and R&D subsidies, the government can incentivize enterprises to innovate in the field of green technology and, in the process, address technological bottlenecks and challenges(Kong et al., 2021; Zhang et al., 2024). Additionally, the government can establish platforms for green technology innovation to

promote the transformation and application of technological achievements. However, the government should promote the development of green industries and build green industrial chains. By formulating and implementing green industrial policies, enterprises can be guided to transition toward green and low-carbon directions. Supporting the development of new energy, new materials, and energy-saving and environmental protection industries can form a complete green industrial ecosystem. Furthermore, through industrial support policies, the government can promote collaborative innovation among upstream and downstream enterprises, jointly advancing the development and application of green technologies and products.

Third, from the perspective of curbing greenwashing behavior and enhancing ESG performance, research indicates that corporate greenwashing behavior and corporate ESG performance are mediating factors through which environmental regulations impact green innovation. Therefore, from a mechanism perspective, the government should take appropriate measures to encourage enterprises to improve ESG performance while curbing greenwashing behavior. The government should formulate and promote corporate ESG assessment standards, encouraging enterprises to voluntarily disclose their ESG performance. By establishing a unified system of ESG assessment, enterprises can more systematically evaluate and enhance their sustainable development capabilities. Additionally, the government can establish ESG awards to recognize and reward outstanding enterprises, setting an example and motivating more companies to improve their ESG performance. The government should simultaneously strengthen its supervision and punitive measures to prevent corporate greenwashing and ensure the seriousness and fairness of environmental regulations. The government can ensure compliance by increasing the scrutiny of corporate environmental data and strictly combating false environmental information. Furthermore, the government should establish and improve the corporate environmental credit system, incorporating greenwashing behavior into the corporate credit evaluation system, deducting credit points, and publicly exposing violators to increase their costs of non-compliance. Reducing the motivation for corporate greenwashing behavior from the root is crucial for the implementation of environmental regulations and the promotion of corporate green innovation.

CRediT authorship contribution statement

Dan Peng: Methodology, Data, Validation, Formal analysis, Visualization, Investigation, Writing – original draft, Writing – review & editing. **Qunxi Kong:** Conceptualization, Supervision, Methodology, Writing – review & editing, Writing – original draft, Funding acquisition.

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Appendix A. Environmental regulation calculation method

This study employs the comprehensive environmental regulation index for the current and the previous period. Drawing on the research of Li and Tao (2012) and Ye et al. (2018), we select unit pollutant emission indicators from prefecture-level cities, including industrial wastewater discharge per unit of output, industrial SO_2 emissions per unit of output, and industrial smoke and dust emissions per unit of output. The entropy method is used to calculate the comprehensive environmental regulation index (*EEFI*); a larger *EEFI* indicates higher pollution emissions, implying weaker environmental regulation and vice versa. The specific steps for calculating the environmental regulation index are as follows:

Step 1: Standardize the data for each indicator as follows:

$$M_{ij} = \frac{X_{ij} - min(X_j)}{max(X_j) - min(X_j)}, \tag{1}$$

where X_{ij} represents the pollutant emissions per unit of output for category j in city i, and M_{ij} is the standardized result; $min(X_j)$ is the minimum value of category j's pollutant emissions across all cities, and $max(X_j)$ is the maximum value.

Step 2: Determine the entropy of each indicator as follows:

$$E_j = -\frac{1}{ln(n)} \sum_{i=1}^{N} p_{ij} lnp_{ij}, \tag{2}$$

where $x'_{ij} = M_{ij}$ and $p_{ij} = \frac{x'_{ij}}{\sum_{i=1}^{n} x'_{ij}}$; p_{ij} represents the proportion of city i under indicator j. If $p_{ij} = 0$ then $lim_{p_{ij} \to 0} p_{ij} lnp_{ij} = 0$.

Step 3: Determine the weights of each indicator as follows:

$$W_j = \frac{1 - E_j}{m - \sum E_j}, \tag{3}$$

where $0 \le j \le m$, and m is the number of prefecture-level cities.

Step 4: Calculate the comprehensive environmental regulation index for city i in year t:

$$EEFI = \sum_{j=1}^{n} W_j M_{ij} \tag{4}$$

Appendix B. Greenwashing behavior measurement method

Corporate Greenwashing Behavior: Domestic and international scholars have conducted exploratory research on the symbolic and substantive actions taken by companies with respect to environmental governance (Walker and Fang, 2012; Roulet and Touboul, 2015; Huang et al., 2019, 2020).

Referencing the studies by Walker and Fang (2012) and Huang et al. (2019, 2020), we construct an indicator system for symbolic and substantive actions in corporate environmental governance with four dimensions: governance and structure, processes and control, inputs and outputs, and compliance and legality (see Table B1). The greenwashing score is calculated using the ratio of the symbolic-action score to the substantive-action score for corporate environmental governance.

Symbolic actions refer to a company's environmental commitments and future plans; these are typically general, abstract statements without substantial implementation or environmental benefits. In contrast, substantive actions are the actual environmental protection measures undertaken by the company that are supported by specific practices and have environmental benefits evidenced by quantifiable data. The detailed calculation process for the greenwashing score is as follows.

First, we employ content analysis to score the items related to corporate environmental disclosure. A value of 1 is assigned if the company has taken substantive action on an item and 0 otherwise. Similarly, a value of 1 is assigned if the company has taken symbolic action on an item and 0 otherwise. During the scoring process, each item is scored by two experts. Formal scoring begins when the consistency between the two scorers reaches 90 % during the trial phase. Any discrepancies between the two experts' scores are resolved by a third expert.

Second, based on the items identified in the first step, the scores for symbolic actions (*GWS*) and substantive actions (*GWR*) are calculated as follows:

$$GWS_{it} = \sum_{i}^{23} x_{ij} \tag{5}$$

$$GWR_{it} = \sum_{i}^{23} x_{ij}, \tag{6}$$

where GWS_{it} is the score for the symbolic actions of company i in year t across all items, and GWR_{it} is the score for the substantive actions of company i in year t across all items; x_{ij} denotes the score of company i for item j ($1 \leq j \leq 23$).

Finally, we calculate the comprehensive greenwashing score. Considering the possibility of zero values in the substantive action scores ($GWR_{it} = 0$), which renders the ratio meaningless, we add 1 to the denominator. The calculation formula is as follows:

$$Greenwash_{it} = \frac{GWS_{it}}{GWR_{it} + 1} \tag{7}$$

Appendix C. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eneco.2024.107971.

References

Chen, Z., Niu, X., Gao, X., et al., 2022. How does environmental regulation affect green innovation? A perspective from the heterogeneity in environmental regulations and pollutants. Front. Energy Res. 10, 885525.

Deng, Y., Wang, L., Zhou, W., 2021. Does environmental regulation promote green innovation capability? - evidence from China. Stat. Res. 38 (07), 76–86.

Du, L., Zhao, Y., Tao, K., et al., 2019. Compound effects of environmental regulation and governance transformation in enhancing green competitiveness. Econ. Res. J. 54 (10), 106–120.

Fang, X., Hu, D., 2023. Corporate ESG performance and innovation: empirical evidence from a-share listed companies. Econ. Res. J. 58 (02), 91–106.

Fang, Z., Kong, X., Şensoy, A., Cui, X., Cheng, F., 2021. Government's awareness of environmental protection and corporate green innovation: a natural experiment from the new environmental protection law in China. Econ. Anal. Policy 70, 294–312.

Flowers, M.E., Matisoff, D.C., Noonan, D.S., 2020. In the LEED: racing to the top in environmental self-regulation. Bus. Strateg. Environ. 29 (6), 2842–2856.

Guo, S., Pei, Y., Wu, Y., 2020. Research on industrial structure adjustment and upgrading effect of the development of productive service industry. J. Quant. Tech. Econ. 37 (10), 45–62.

Hering, L., Poncet, S., 2014. Environmental policy and exports: evidence from Chinese cities. J. Environ. Econ. Manag. 68 (2), 296–318.

Huang, R., Chen, W., Wang, K., 2019. External financing demand, impression management, and enterprise greenwashing. Comp. Econ. Soc. Syst. 03, 81–93.

Huang, R., Xie, X., Zhou, H., 2020. 'Isomorphic' behavior of corporate greenwashing. China Popul. Resour. Environ. 30 (11), 139–150.

Jong, M., Harkink, K., Barth, S., 2017. Making green stuff? Effects of corporate greenwashing on consumers. J. Bus. Tech. Commun. 32, 77–112.

Kolcava, D., 2022. Greenwashing and public demand for government regulation. J. Public Policy 43, 179–198.

Li, L., Tao, F., 2012. Selection of optimal environmental regulation intensity for Chinese manufacturing industry—based on the green TFP perspective. China Ind. Econ. 05, 70–82.

Li, X., Hu, Z., Zhang, Q., 2021. Environmental regulation, economic policy uncertainty, and green technology innovation. Clean Techn. Environ. Policy 23, 2975–2988.

Kong, Q., Li, R., Wang, Z., Peng, D., 2022. Economic policy uncertainty and firm investment decisions: Dilemma or opportunity? International Review of Financial Analysis 83, 102301.

Kong, Q., Peng, D., Ni, Y., Jiang, X., Wang, Z., 2021. Trade openness and economic growth quality of China: Empirical analysis using ARDL model. Finance Research Letters 38, 101488.

Li, W., Bhutto, M.Y., Waris, I., et al., 2023. The nexus between environmental corporate social responsibility, green intellectual capital and green innovation towards business sustainability: an empirical analysis of Chinese automobile manufacturing firms. Int. J. Environ. Res. Public Health 20 (3), 1851.

Peng, D., Ji, Y., Kong, Q., 2023. OFDI and firms' sustainable productive capacity: Evidence from Chinese industrial firms. International Review of Economics & Finance 83, 641–652.

Peng, H., Shen, N., Ying, H., Wang, Q., 2021. Can environmental regulation directly promote green innovation behavior?——based on situation of industrial agglomeration. J. Clean. Prod. 314, 128044.

Porter, M.E., Linde, C.V.D., 1995. Toward a new conception of the environment-competitiveness relationship. J. Econ. Perspect. 9 (4), 97–118.

Qi, S., Lin, S., Cui, J., 2018. Do environmental rights trading schemes induce green innovation? Evidence from listed firms in China. Econ. Res. J. 53 (12), 129–143.

Roulet, T.J., Touboul, S., 2015. The intentions with which the road is paved: attitudes to liberalism as determinants of greenwashing. J. Bus. Ethics 128 (2), 305–320.

Shi, H., Yi, M., 2020. Research on the influence of environmental regulation on high-quality development and spatial spillover effect. Inq. Econ. Issues 05, 160–175.

Walker, K., Fang, W., 2012. The harm of symbolic actions and green-washing: corporate actions and communications on environmental performance and their financial implications. J. Bus. Ethics 109 (2), 227–242.

Wang, X., Wang, Y., 2021. Research on the green innovation promoted by green credit policies. J. Manag. World 37 (06), 173-188+11.

Wang, D., Walker, T., Barabanov, S., 2020. A psychological approach to regaining consumer trust after greenwashing: the case of Chinese green consumers. J. Consum. Mark. 37, 593–603.

Wang, Z., Li, R., Kong, Q., 2024. Shanghai oil futures prices and firms' investment decisions: Evidence from Chinese manufacturing firms. Energy Economics 134, 107560.

Wang, D., Luo, Y., Hu, S., Yang, Q., 2022a. Executives' ESG cognition and enterprise green innovation: evidence based on executives' personal microblogs. Front. Psychol. 13.

Wang, M., Zhou, J., Xia, X., Wang, Z., 2022b. The mixed impact of environmental regulations and external financing constraints on green technological innovation of Enterprise. Int. J. Environ. Res. Public Health 19.

Wang, F., He, J., Chen, L., 2023. Will interlocking directors with green experience promote quantity increase and quality improvement of enterprise green innovation. China Ind. Econ. 10, 155–173.
Wen, Z., Ye, B., 2014. Analyses of mediating effects: the development of methods and models. Adv. Psychol. Sci. 22 (05), 731–745.
Xie, H., Lyu, X., 2022. Responsible multinational investment: ESG and Chinese OFDI. Econ. Res. J. 57 (03), 83–99.
Xu, Y., 2014. Whether informal environmental regulation from social pressure constraints on China's industrial pollution? Financ. Trade Res. 25 (02), 7–15.
Ye, Q., Zeng, G., Dai, S., et al., 2018. Research on the effects of different policy tools on China's emissions reduction innovation: based on the panel data of 285 prefectural-level municipalities. China Popul. Resour. Environ. 28 (02), 115–122.
York, R., Rosa, E.A., 2003. Key challenges to ecological modernization theory: institutional efficacy, case study evidence, units of analysis, and the pace of eco-efficiency. Organ. Environ. 16 (3), 273–288.
Zhang, D, Bai, D, Chen, X, 2024. Can crude oil futures market volatility motivate peer firms in competing ESG performance? An exploration of Shanghai International Energy Exchange[J]. Energy Economics 129, 107240.
Zhang, D, Bai, D, Wang, C, et al., 2024. Distribution dynamics and quantile dynamic convergence of the digital economy: Prefecture-level evidence in China[J]. International Review of Financial Analysis 95, 103345.
Zhang, D., Bai, D., Wang, Y., 2024. Green vs. brown: Climate risk showdown–who's thriving, who's diving? Journal of International Money and Finance 103198.
Zhang, D, He, Y, Lu, M, 2024. Is energy firms' investment behavior more sensitive on corporate perception of monetary policy? Energy Economics 136, 107749.
Zhang, D., Kong, Q., 2022. Green energy transition and sustainable development of energy firms: An assessment of renewable energy policy. Energy Economics 111, 106060.
Zhang, Y., Wang, Y., 2022. Do managerial ties help or hinder corporate green innovation? The moderating roles of contextual factors. Int. J. Environ. Res. Public Health 19.
Zhang, D., Wang, C., Yurun, H.E., Vigne, S.A., 2024. Does FinTech efficiently hamper manipulating ESG data behavior? The British Accounting Review 101494.